

Machine Learning for Natural Language Understanding

Prof. Achim Rettinger

Language Models

**Cats Eyes
Removed**

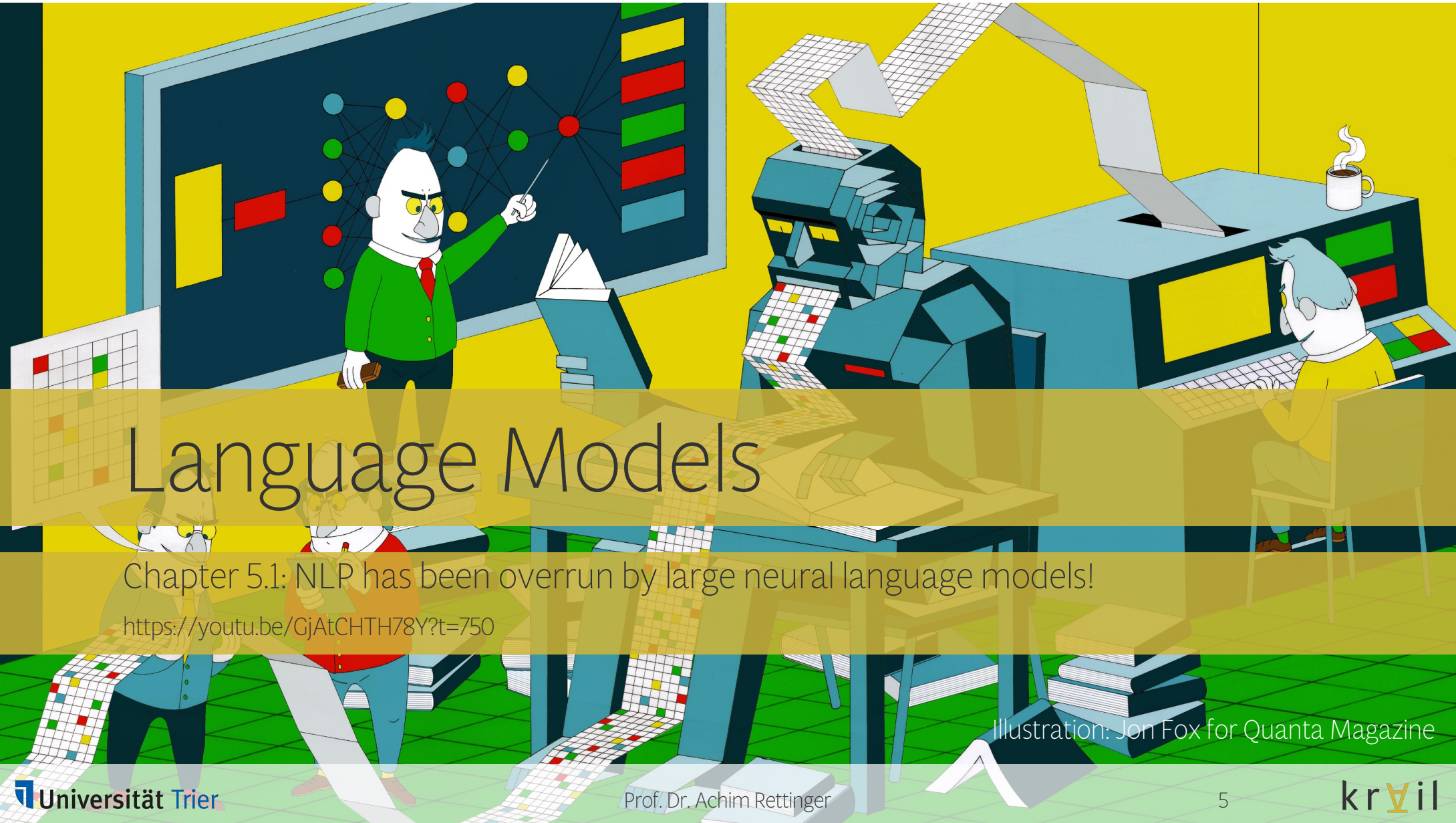
Lecture 5 – Language Models 1

Sources Lecture and Exercise

- Language Models and RNNs
 - <https://youtu.be/PLryWeHPcBs?t=3000> (Intro and RNN basics)
 - <https://youtu.be/OLixFSa7yts> (starts with training RNNs)
 - <https://youtu.be/OLixFSa7yts?t=1091> (Generating text with RNNs)
 - <https://youtu.be/OLixFSa7yts?t=1459> (Evaluating Language Models (Perplexity))
 - <https://youtu.be/OLixFSa7yts?t=1772> (Applications of LMs)
 - <https://youtu.be/OLixFSa7yts?t=2218> (Issues (Vanishing Gradient))
 - <https://youtu.be/OLixFSa7yts?t=2983> (LSTMs)
 - <https://youtu.be/OLixFSa7yts?t=4596> (Bi-LSTMs and Deep LSTMs)
- Attention in RNNs
 - <https://youtu.be/wzfWHP6SXxY?t=4367> (Seq2Seq tasks)

Flipped Classroom Questions

- What is a “neural language model”? What is the advantage over n-grams.
- What is the difference between contextualized word embeddings vs static word embeddings (like e.g., from word2vec)?
- How do RNNs overcome the limited window size?
- How many gates are in an LSTM?



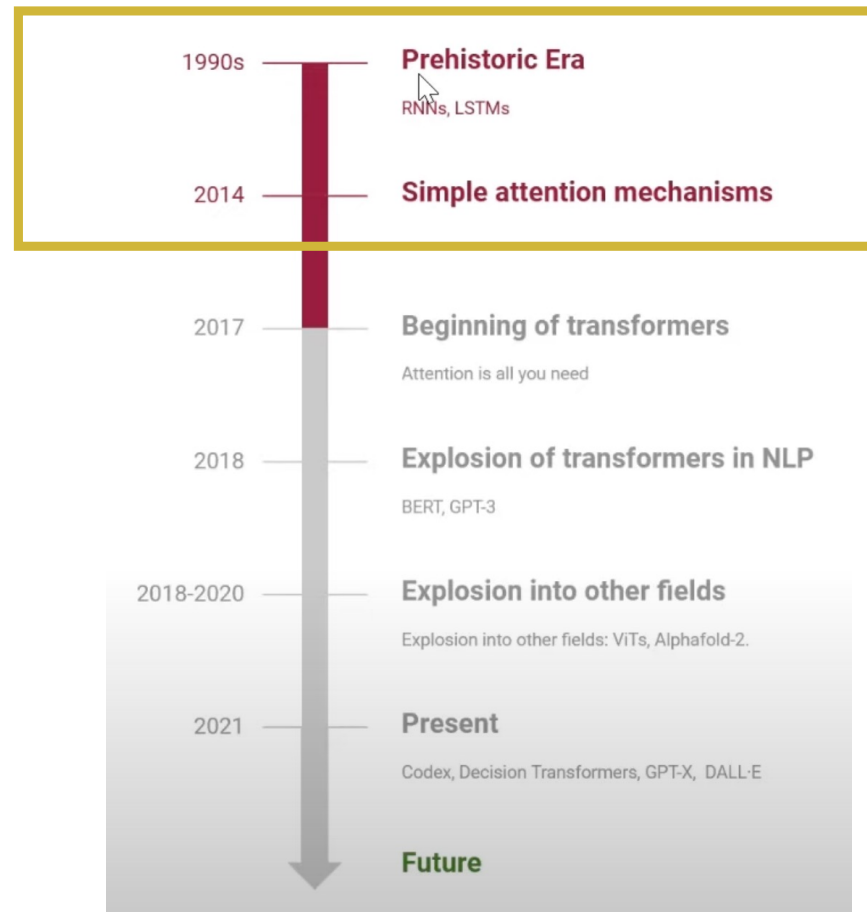
Language Models

Chapter 5.1: NLP has been overrun by large neural language models!

<https://youtu.be/CjAtCHTH78Y?t=750>

Illustration: Jon Fox for Quanta Magazine

Historic timeline – Focus of lecture today



<https://youtu.be/P127jhj-8-Y>

Structure vs Frequency

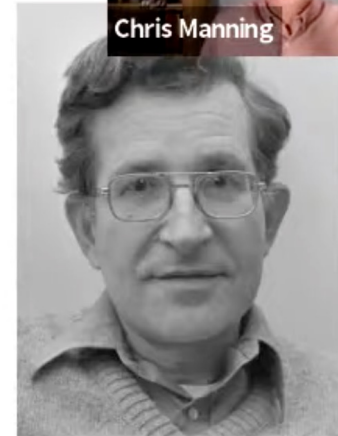
Noam Chomsky vs. Claude Shannon

Chomsky (1969):

“But it must be recognized that the notion of ‘probability of a sentence’ is an entirely useless one, under any known interpretation of this term”

Shannon (1951):

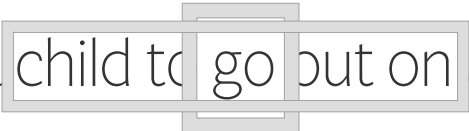
“anyone speaking a language possesses, implicitly, an enormous knowledge of the statistics of the language [enabling them] to complete an unfinished phrase”



Recap: Word Window

- Center word vs context words

I loved when I was a child to go out on a ranch...

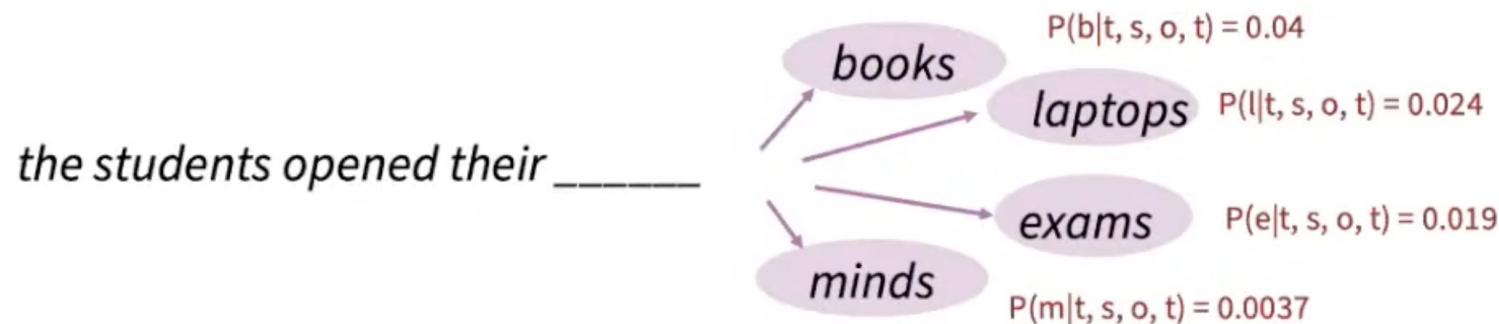
A diagram illustrating a word window. The sentence "I loved when I was a child to go out on a ranch..." is shown. A gray rectangular box highlights the words "to go out". Within this box, a smaller white rectangular box highlights the word "go". This visualizes the concept of a center word (w_t) and its context words (w_{t-j} to w_{t+j}).

$$p(w_{t-j}|w_t) \quad w_t \quad p(w_{t+j}|w_t)$$



Language Modeling

A **Language Model (LM)** predicts a word in a context



LMs in The Dark Ages (20th C): *n*-gram model

Count how often words follow word sequences; divide to get cond. prob.

$$\#(t, s, o, t, b) = 11 \quad \#(t, s, o, t) = 187 \quad P(b | t, s, o, t) = 11/187 = 0.06$$

Before 2000: N-Grams

N-gram models almost work as n increases



1-gram:

To him swallowed confess hear both. Which. Of save on trail for are ...

2-gram:

Why dost stand forth thy canopy, forsooth; he is this palpable hit the King

3-gram:

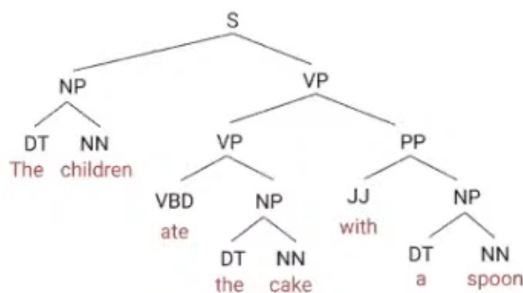
Fly, and will rid me these news of price. Therefore the sadness of parting,

2010: Recurrent Neural Networks

Recurrent neural networks and human language syntax

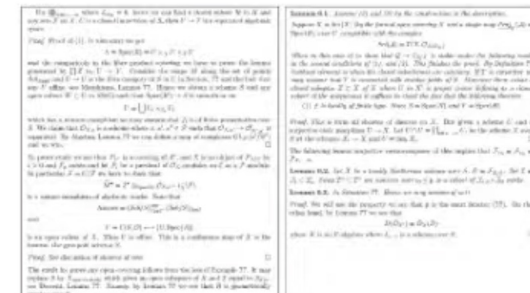


Chris Manning



agree in number

The **chef** who made the **pizzas** **is** **are**

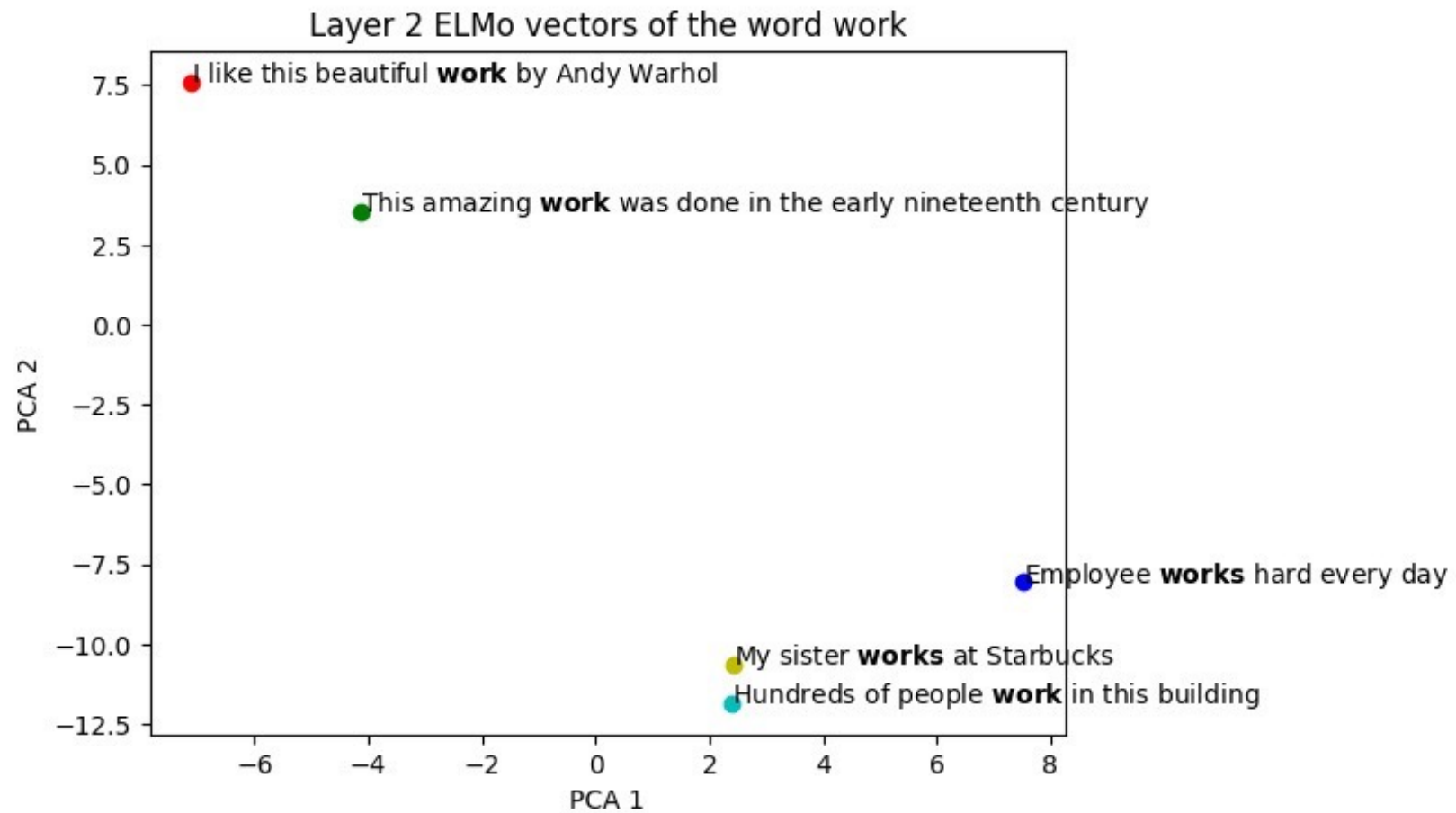


Human languages like English exhibit hierarchical or recursive structure, often with nested dependencies

RNN sequence models perform well on syntax-dependent language model predictions like agreement ¹

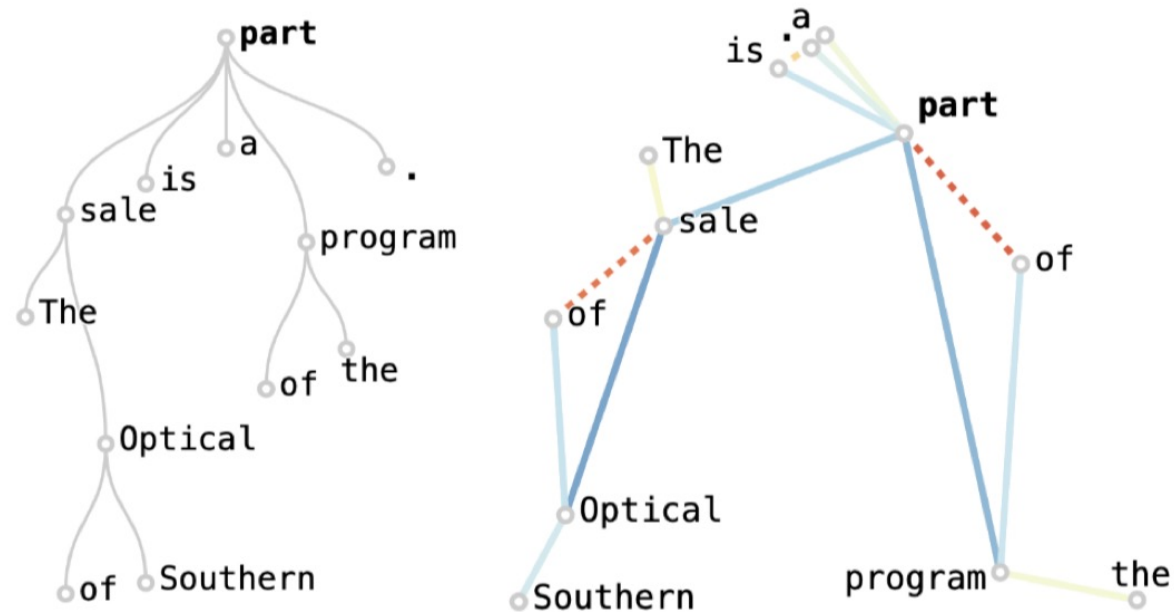
RNNs' generations can model hierarchical syntax, like this LaTeX from *The Unreasonable Effectiveness of RNNs*²

2013: LSTMs



Since 2018: Transformers

“The sale of Southern Optical is a part of the program.”



<https://arxiv.org/pdf/1906.02715v1.pdf>

Degrees of contextualized word representations

- Sequence-ignorant:
 - Bag-of-words inside word window
- Relative-positional feature
 - Relative position of context words
- Directional features
 - All previous and/or following words
- All words within the sentence
 - with absolute positions (see next lecture)

Relative-positional: N-Grams

Chapter 5.2

N-Grams

- Letter/character ngrams: one/bi/tri/four/five
- Word-bigrams, trigrams
- Bag-of-bigrams
- Typically fixed-length

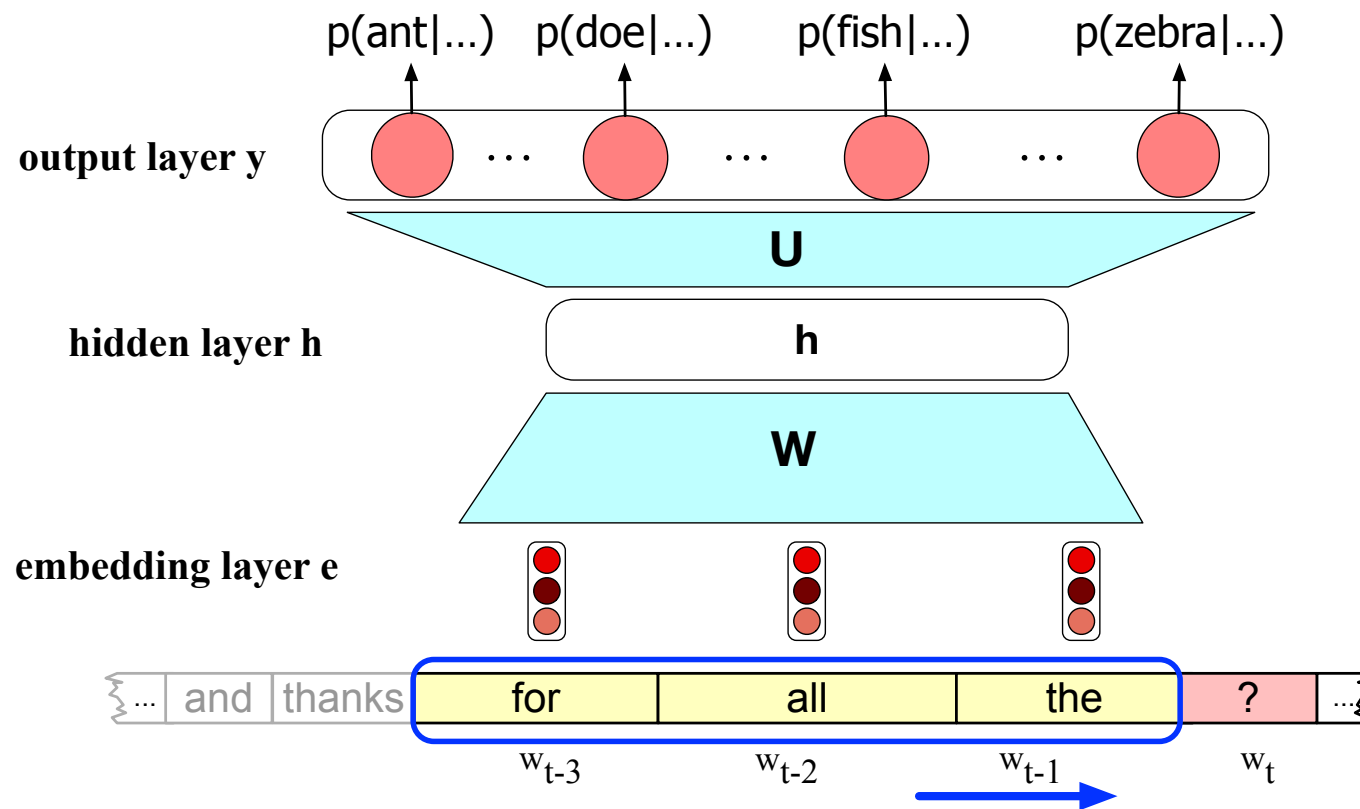
The logo for Google Books Ngram Viewer, featuring the word "Google" in its multi-colored font, followed by "Books" in grey and "Ngram Viewer" in green.

<https://books.google.com/ngrams>

Directional Features: RNNs

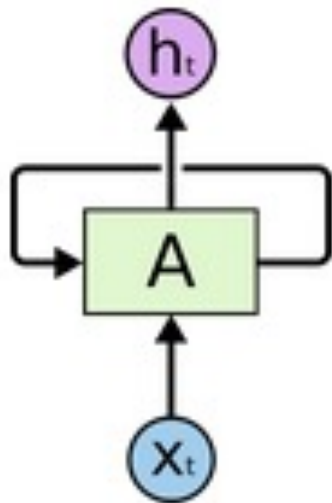
Chapter 5.3

Naïve MLP for Next Word Prediction



<https://web.stanford.edu/~jurafsky/slp3/9.pdf>

Recurrent Neural Network (RNN)



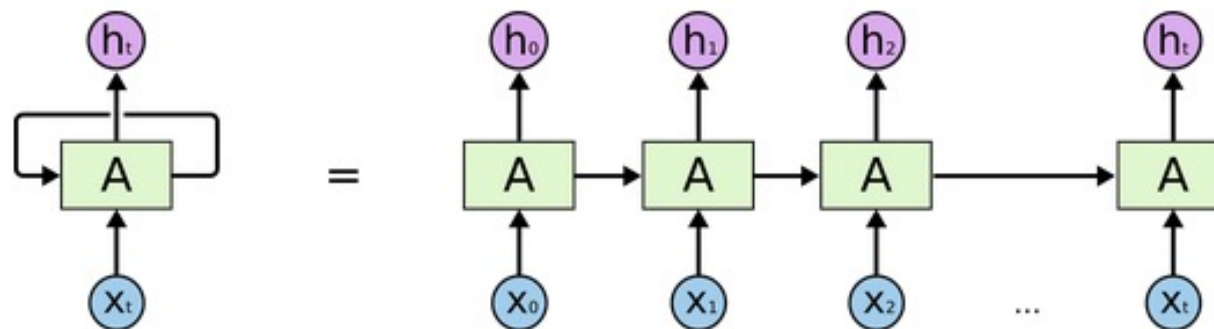
Recurrent Neural Networks have loops

A:

- looks at some input x_t
- ... and outputs a value h_t

<http://colah.github.io/posts/2015-08-Understanding-LSTMs/>

Unrolled RNN

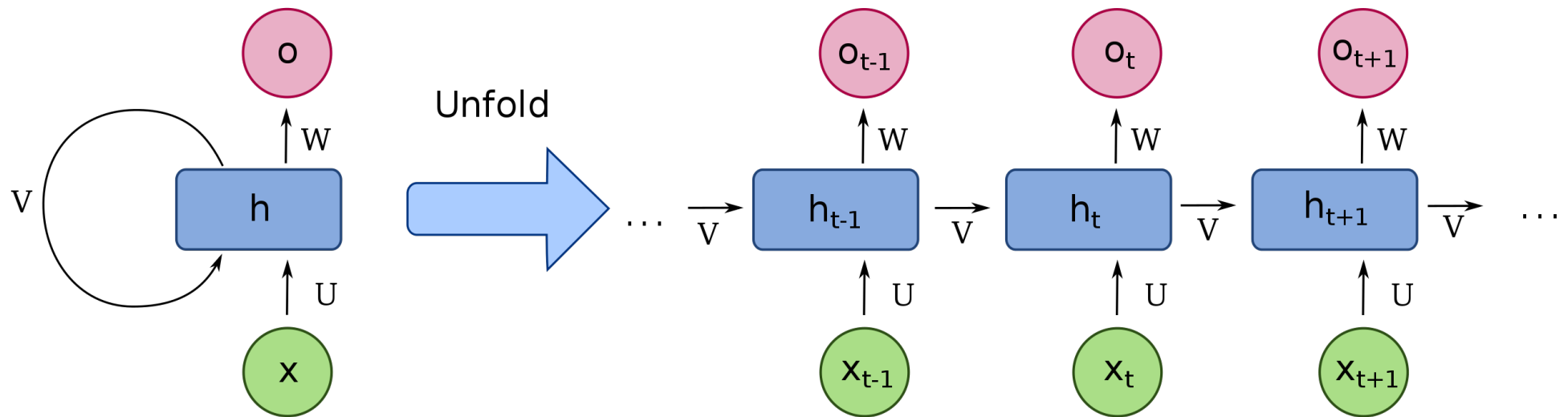


A class of artificial neural networks where connections between nodes form a directed graph along a temporal sequence.

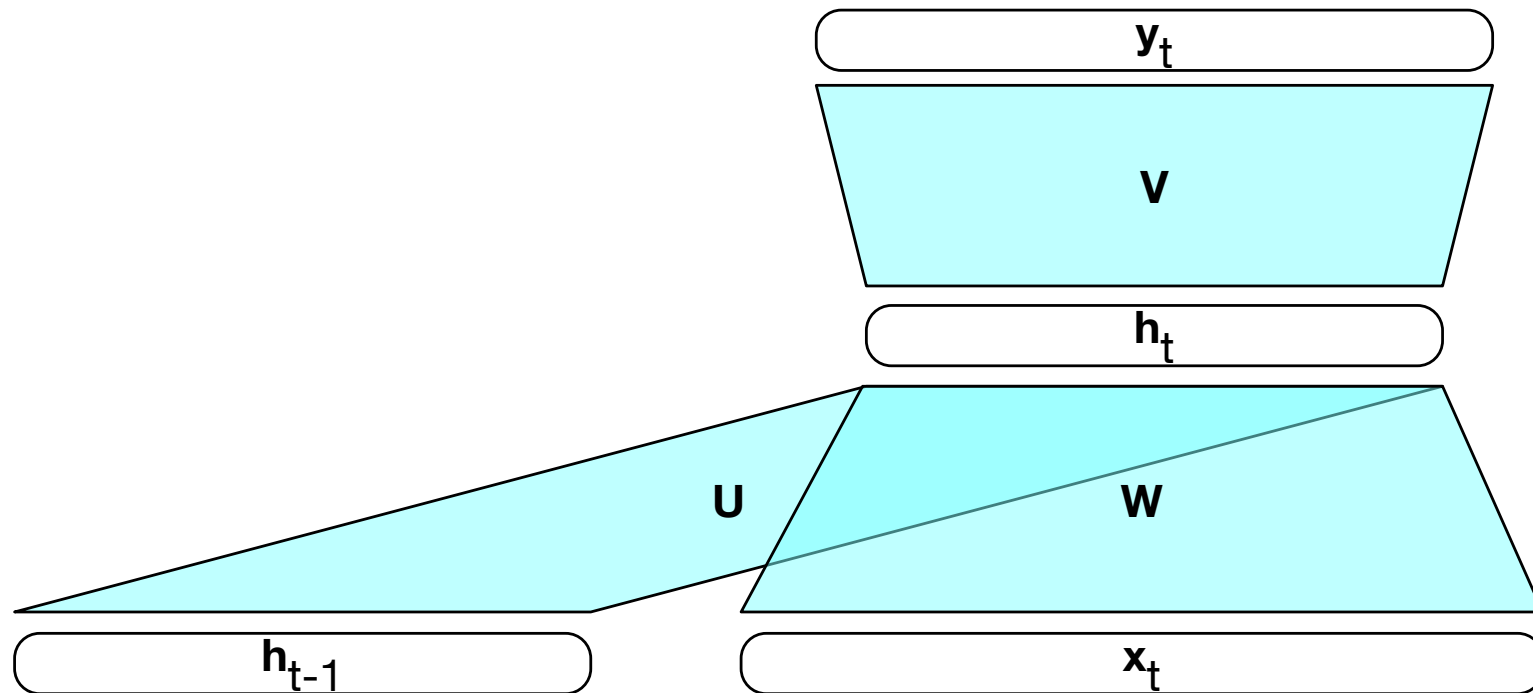
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Recurrent Neural Networks

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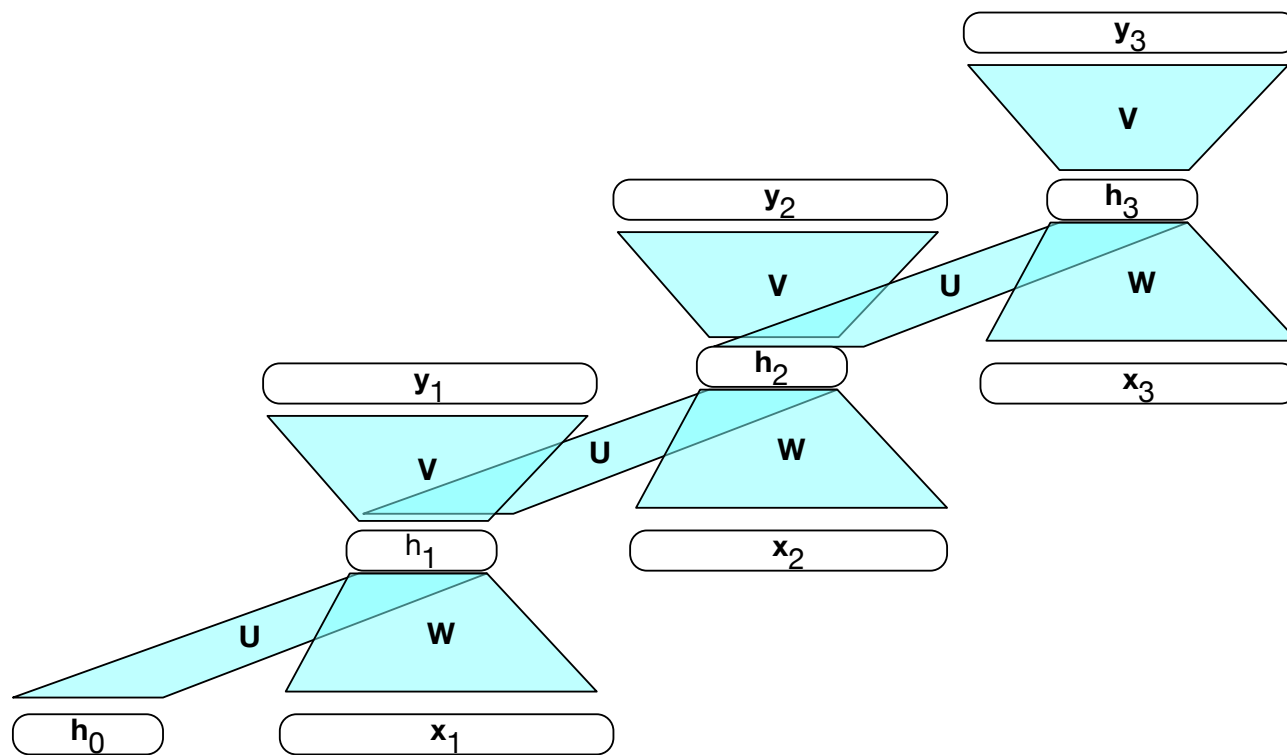


One-step RNN



<https://web.stanford.edu/~jurafsky/slp3/9.pdf>

Multiple-step RNN

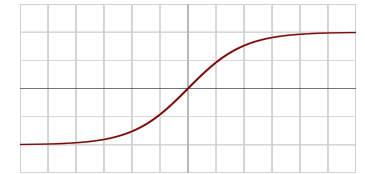
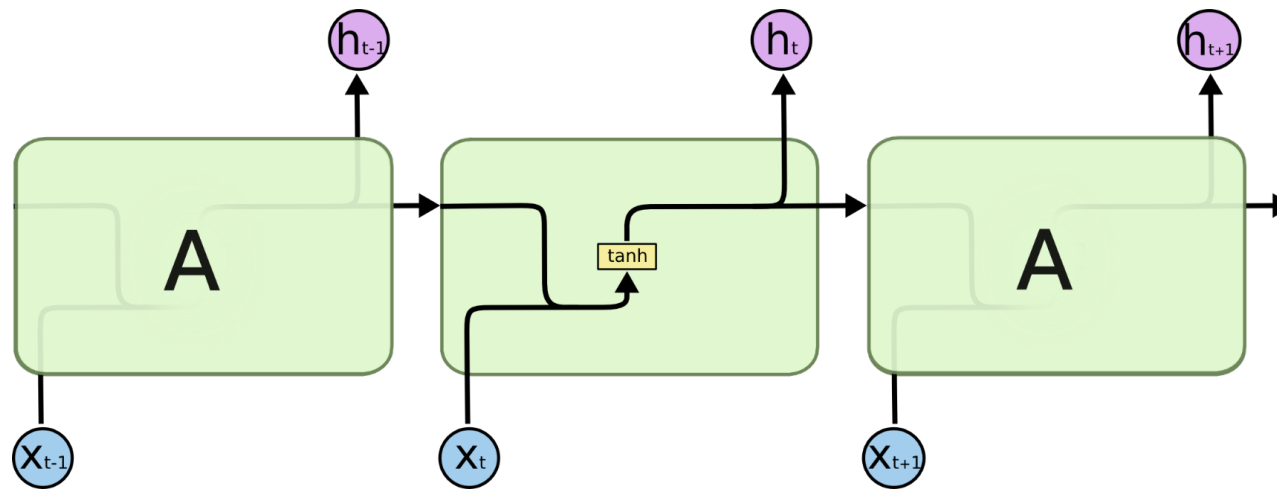


<https://web.stanford.edu/~jurafsky/slp3/9.pdf>

Sophisticated RNNs: LSTM

Chapter 5.4

Standard RNN (e.g. with tanh activation function)

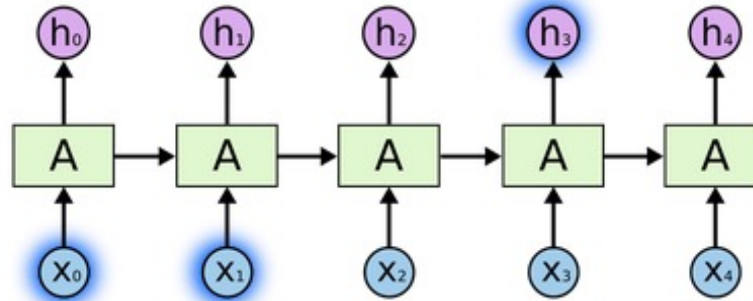


https://en.wikipedia.org/wiki/Activation_function

<http://colah.github.io/posts/2015-08-Understanding-LSTMs/>

Shortcomings of RNNS

Problem of Long-Term Dependencies

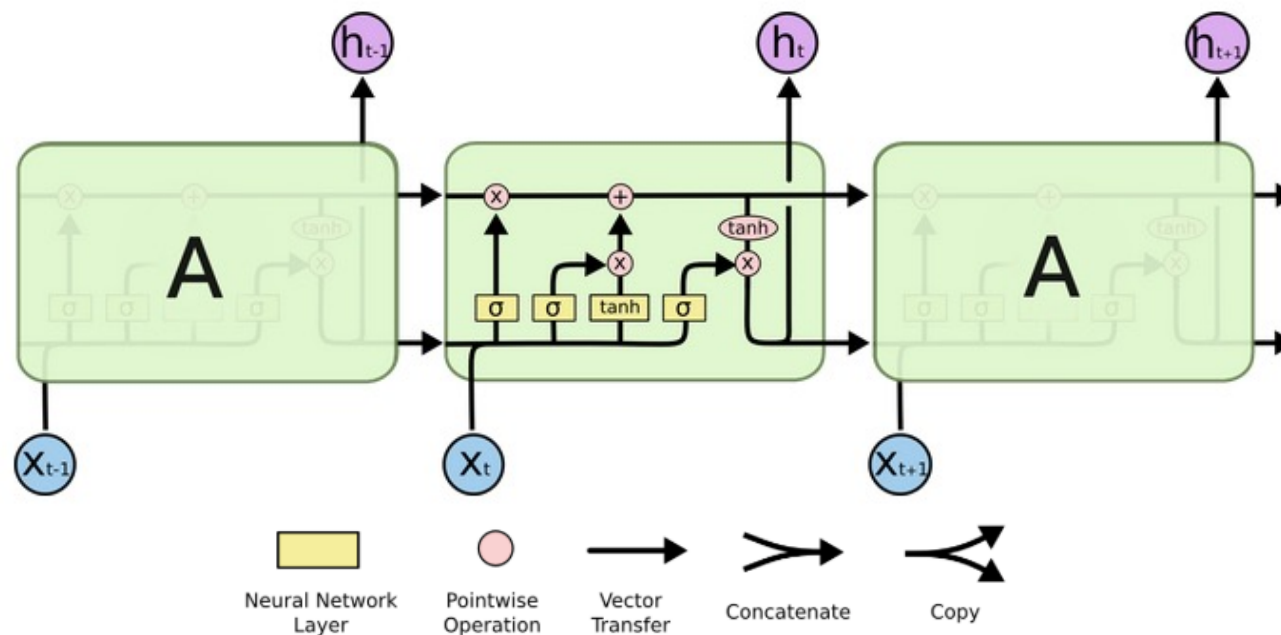


RNN do not capture long-term dependencies and fails to keep the context.

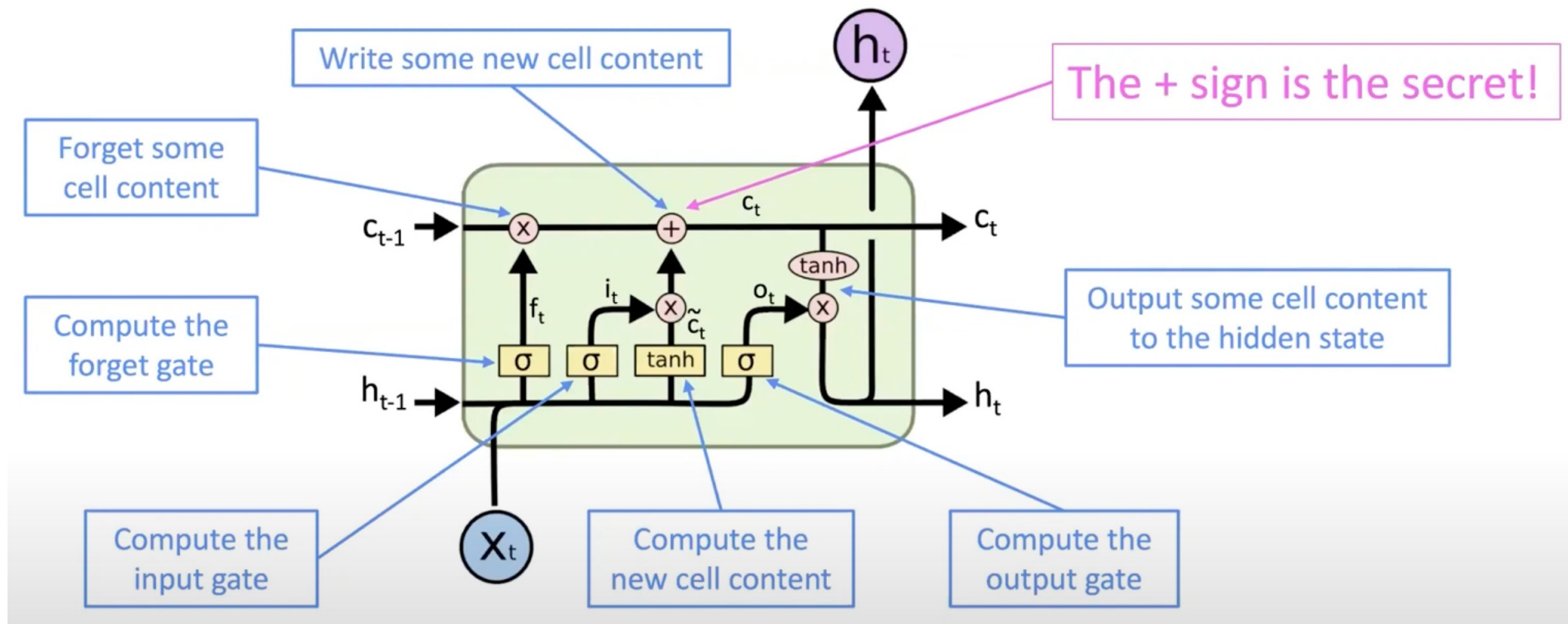
<http://colah.github.io/posts/2015-08-Understanding-LSTMs/>

Long Short-Term Memory (LSTM) I

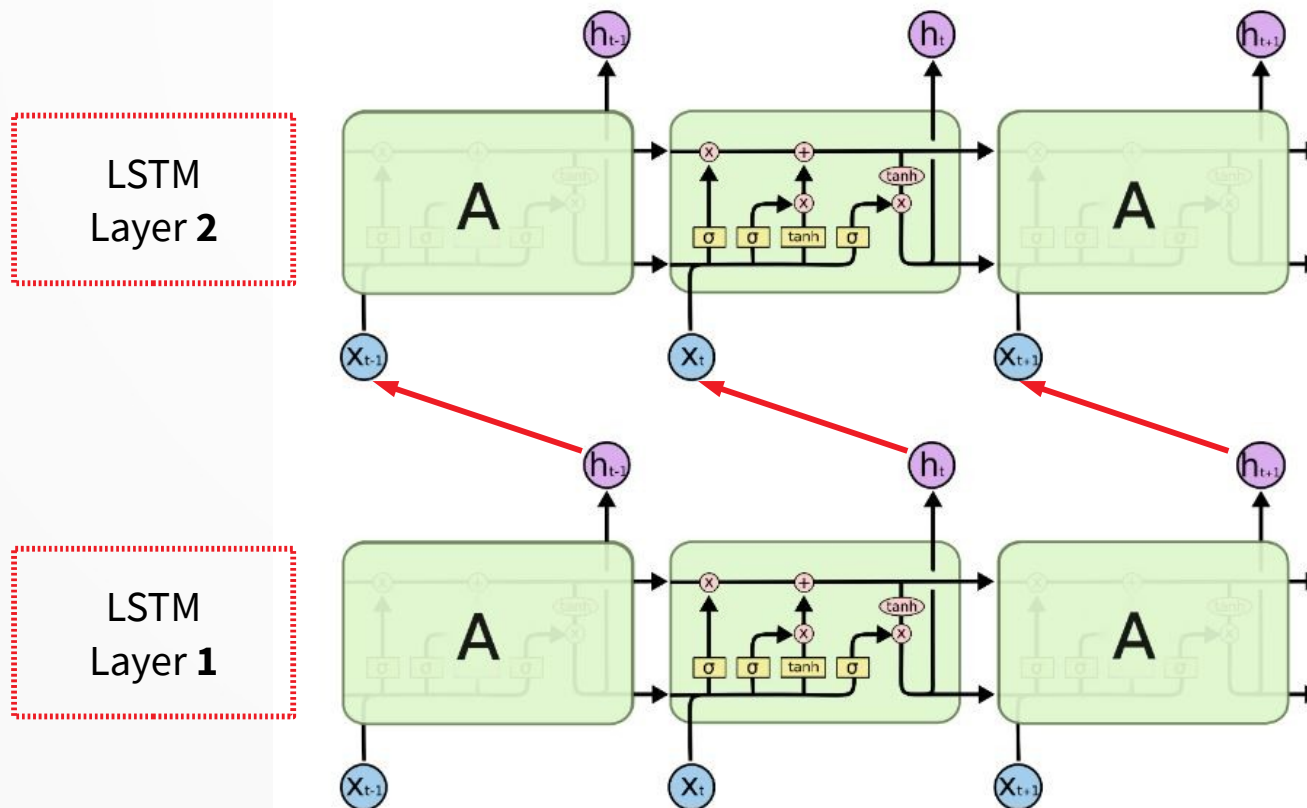
Core idea: add “cell state” (= additional memory)



Long Short-Term Memory (LSTM) II



Deep LSTMs

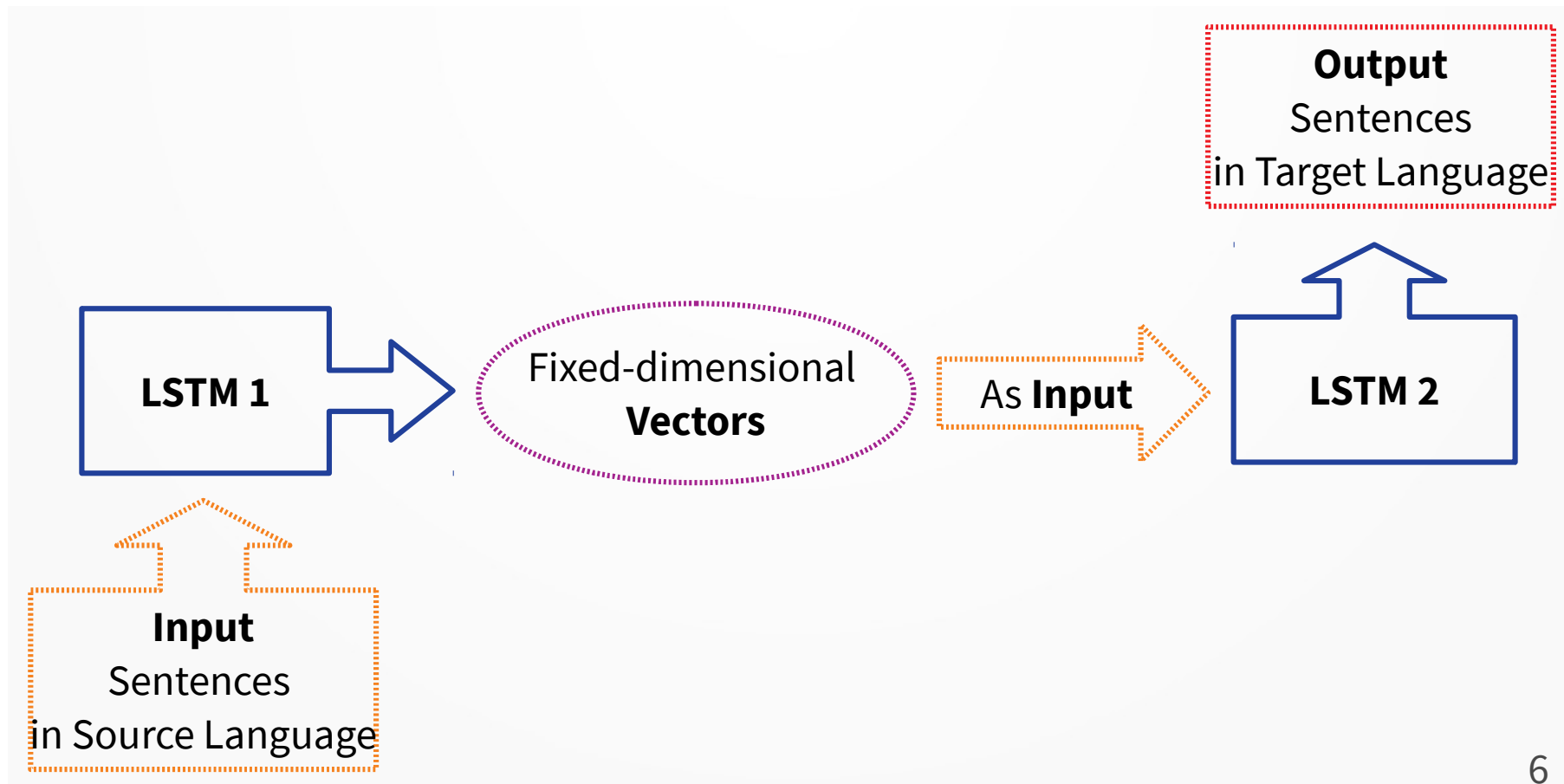


Source: <https://colah.github.io/posts/2015-08-Understanding-LSTMs/>

Application: Sequence to Sequence

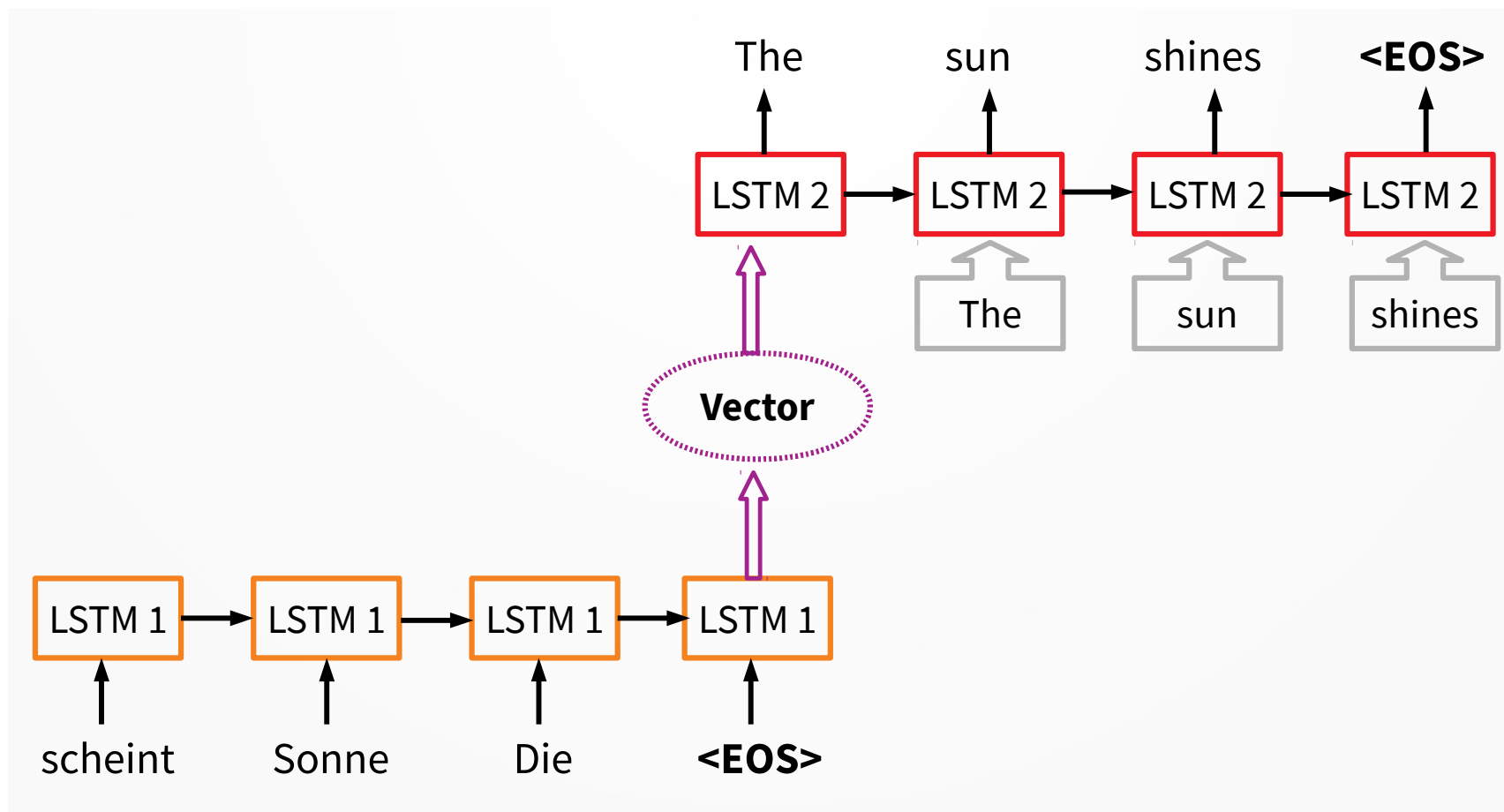
Chapter 5.5

Sequence to Sequence Models



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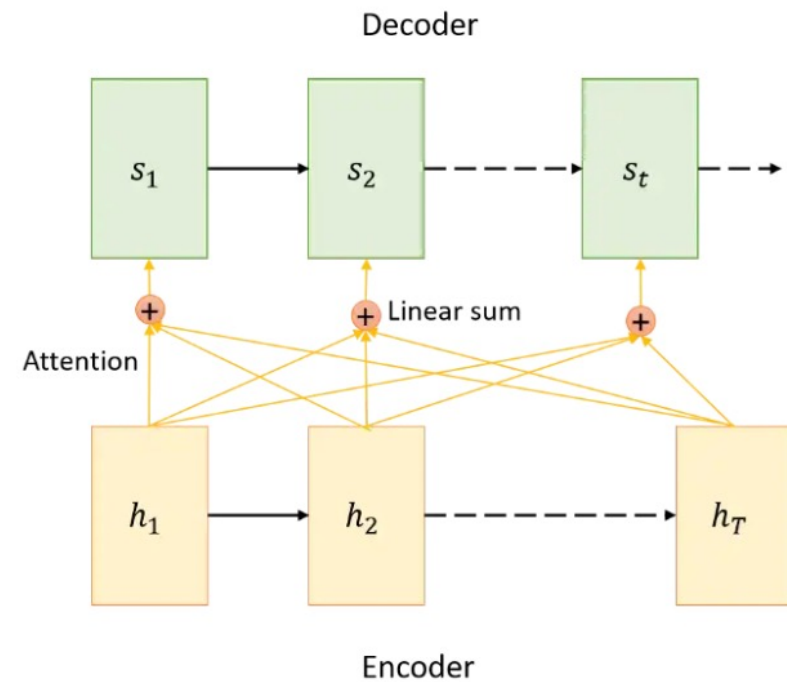
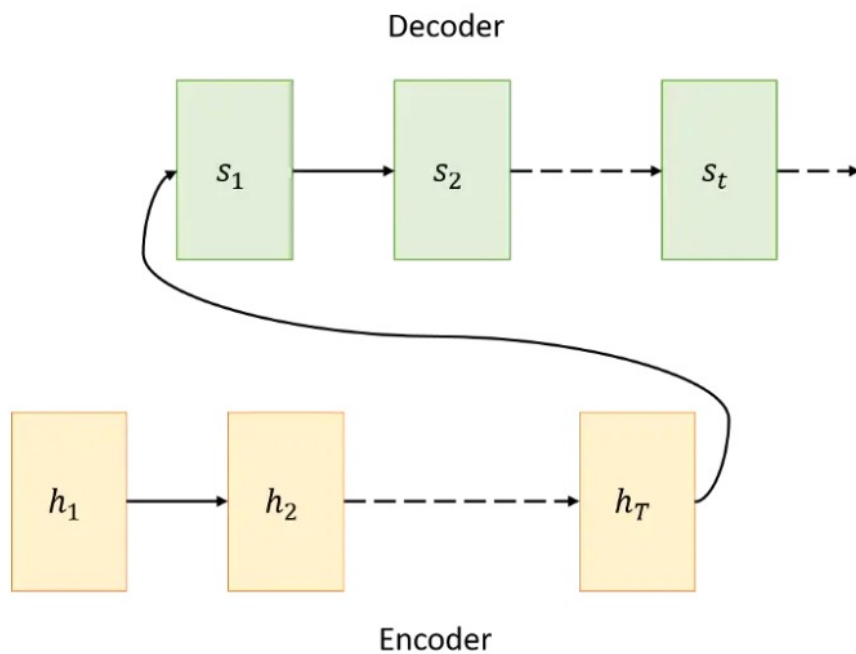
Machine Translation with LSTMs



Attention in RNNs

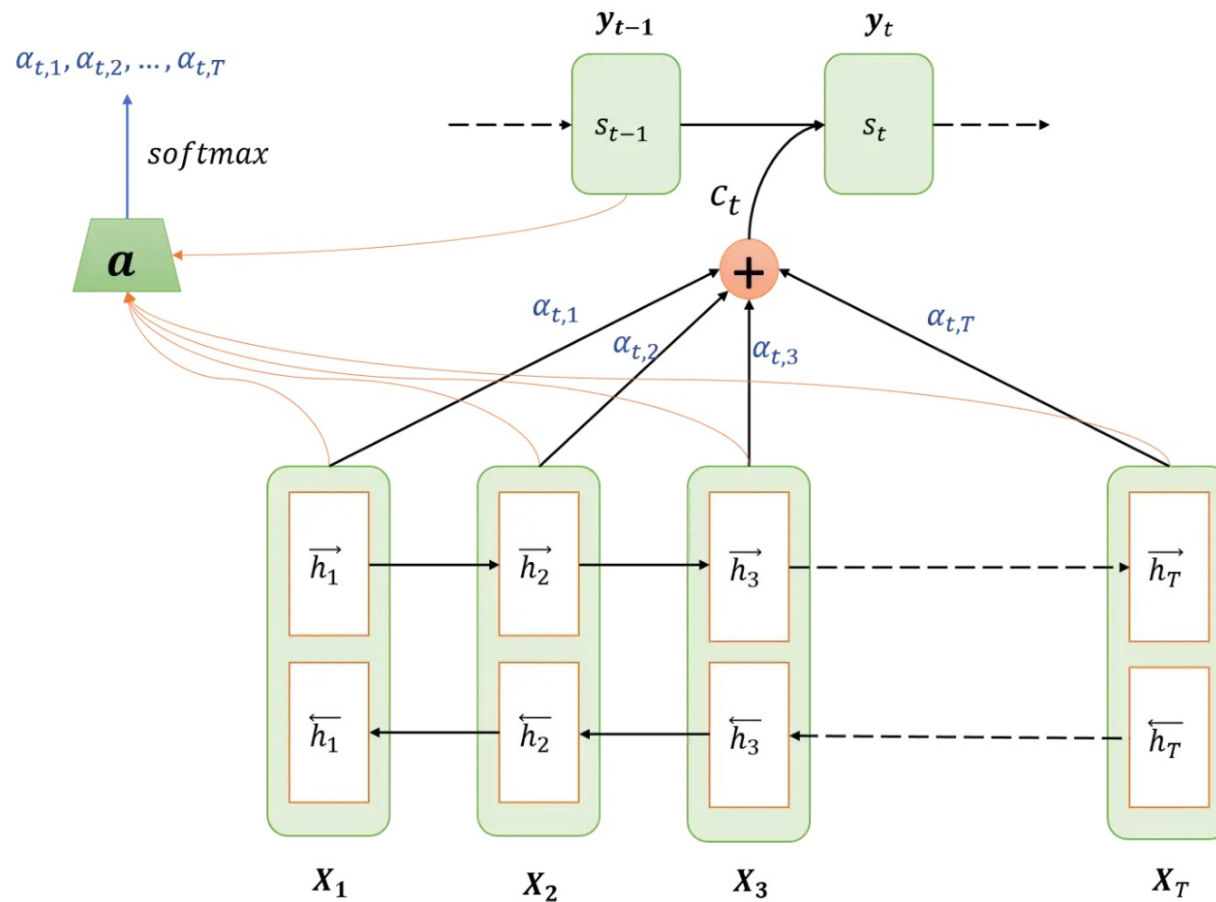
Chapter 5.6

Simple vs Attention based encoder-decoder



<https://medium.com/swlh/a-simple-overview-of-rnn-lstm-and-attention-mechanism-9e844763d07b>

Attention Mechanism architecture overview



“The future of language is about the development of language models...”¹

1

Computerlinguistik | Universität Trier | cl.cldh.uni-trier.de